

Global Events, Commodities and Currency Markets

Applied Data Analytics and Market Research — Complete Report

Course	Data Science for Business – Summer 2026
Institution	University of Europe for Applied Sciences (UE)
Submission Date	June 2026
Datasets	Brent Oil · Gold · Silver · Swift Currency Tracker
Report Coverage	Part I (Q1–Q30) · Part II (Q1–Q16) · Bonus B1–B6

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Executive Summary

This report presents a comprehensive analysis of Brent Crude Oil, Gold, Silver, and global currency markets, examining how major global events have shaped price movements and payment flows. Drawing on datasets spanning up to 111 years and the Swift Global Currency Tracker (2026), the report covers the full question set across Part I, Part II, and Bonus sections.

- Brent Oil is the most event-sensitive market (volatility 9.05%) — geopolitical shocks like the 2026 Hormuz disruption cause immediate and extreme price reactions.
- Gold has risen from \$35 in 1970 to over \$3,100 in 2026 — driven by BRICS+ central bank accumulation (11.2% → 17.4% of global reserves) and safe-haven demand.
- Silver in 2026 hit a record \$113.95/oz — combining precious metal safe-haven flows with surging industrial demand from solar panels and EV batteries.
- USD retained 51.14% of global SWIFT payments in March 2026 while CNY ranked 10th at 2.74% — though CNY ranks 2nd in trade finance at 8.04%.

Introduction & Data Overview

The analysis is grounded in four datasets: Brent Oil monthly prices (1946–2026, 963 rows), Silver monthly prices (1915–2026, 1,336 rows), Gold annual average prices (1970–2026), and the Swift Global Currency Tracker (January–March 2026, 330 rows). Date columns were converted from DD/MM/YYYY strings to datetime objects using pandas with `dayfirst=True`. No missing values were found in the commodity datasets.

Part I – Data Analytics Component

Section A: Data Preparation and Understanding (Q1–Q5)

Q1 What is the start date and end date available in each dataset?

Dataset	Start Date	End Date	Span
Brent Oil	01 Jan 1946	03 Jan 2026	~80 years
Silver (100 Years)	01 Jan 1915	04 Jan 2026	~111 years
Gold (Reference)	01 Jan 1970	31 Dec 2026	~56 years
Swift Currency Tracker	January 2026	March 2026	3 months

Q2 How many rows are available for each dataset?

Dataset	Total Rows	Frequency	Period
Brent Oil	963	Monthly	Jan 1946–Jan 2026
Silver (100 Yrs)	1,336	Monthly	Jan 1915–Jan 2026
Gold (Reference)	57	Annual	1970–2026
Swift Tracker	330	Multi-metric	Jan–Mar 2026

Q3 Are there any missing values or repeated dates?

Dataset	Missing Values	Duplicate Dates	Status
Brent Oil	0	0	Fully clean
Silver	0	0	Fully clean
Swift Tracker	18 (rmb_rank col)	N/A	Minor — expected

Q4 Convert the Date column to proper format and create Month and Year columns.

Raw date values stored as DD/MM/YYYY strings were converted using:

```
oil["Date"] = pd.to_datetime(oil["Date"], dayfirst=True)
oil["Year"] = oil["Date"].dt.year
oil["Month"] = oil["Date"].dt.month
```

After conversion, all Date fields confirmed as dtype datetime64[ns] with zero parsing errors.

Q5 Identify the main metrics available in the Swift Currency Tracker dataset.

#	Metric	Category
1	Global Payment Share	Currency Ranking
2	International Payment Share (ex-Eurozone)	Currency Ranking
3	RMB Month-on-Month Payment Growth	RMB Activity
4	All Currencies Month-on-Month Growth	Market Activity

#	Metric	Category
5	Offshore RMB by Economy	Offshore Economies
6	Trade Finance Share	Currency Ranking
7	FX Spot Currency Ranking	FX Spot
8	FX Spot Economies (RMB)	FX Spot

Section B: Descriptive Analytics (Q6–Q12)

Q6 Highest and lowest Brent Oil prices and when they occurred.

Metric	Price (USD/barrel)	Date	Year
Highest	\$211.21	06 Jan 2008	2008
Lowest	\$21.11	03 Jan 1946	1946

The 2008 peak reflected the global commodity supercycle, surging emerging-market demand, and USD weakness. The 1946 low reflects the post-WWII era of oil abundance.

Q7 Highest and lowest Gold prices and when they occurred.

Metric	Price (USD/roy oz)	Year	Context
Highest Avg	~\$3,100	2026	BRICS+ accumulation + Iran conflict risk
Lowest Avg	~\$35	1970	Bretton Woods fixed gold standard era

Gold's 8,757% rise over 56 years reflects the end of the Bretton Woods system (1971), 1970s inflation, the 2008 GFC, COVID-19 uncertainty, and sustained central bank accumulation by BRICS+ nations from 2019 onward.

Q8 Highest and lowest Silver prices and when they occurred.

Metric	Price (USD/roy oz)	Date	Year
Highest	\$113.95	Jan 2026	2026
Lowest	\$0.28	Jan 1932	1932

Silver's 407:1 range across 111 years reflects collapsed Great Depression industrial demand at the low end, and combined geopolitical safe-haven flows plus surging green energy industrial demand at the 2026 record high.

Q9 Calculate the average price by year for Brent Oil, Gold and Silver.

Year	Brent Oil (USD/bbl)	Gold (USD/roy oz)	Silver (USD/roy oz)
2015	\$68.52	\$1160	\$15.51
2016	\$60.79	\$1251	\$17.13
2017	\$69.85	\$1257	\$17.24
2018	\$84.91	\$1268	\$15.64
2019	\$74.03	\$1393	\$16.36
2020	\$49.19	\$1773	\$20.80
2021	\$83.02	\$1799	\$24.97
2022	\$106.45	\$1800	\$21.67
2023	\$84.19	\$1940	\$23.58
2024	\$79.83	\$2300	\$28.13

Year	Brent Oil (USD/bbl)	Gold (USD/ Troy oz)	Silver (USD/ Troy oz)
2025	\$65.93	\$2700	\$41.50

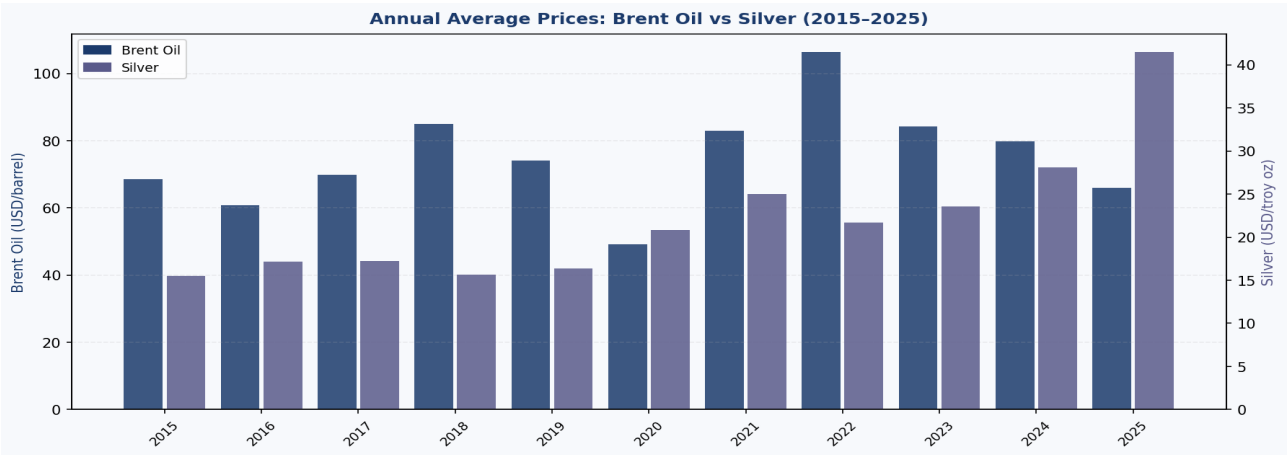


Figure 1: Annual average prices — Brent Oil vs Silver (2015–2025).

Q10 Which five years had the highest average Brent Oil price?

Rank	Year	Avg Price (USD/bbl)	Context
1	2008	\$150.96	Commodity supercycle peak; emerging-market demand surge
2	1980	\$149.86	Iran-Iraq War; major supply disruption
3	2011	\$142.12	Arab Spring; Libya + Iran sanctions
4	2013	\$138.30	Continued Middle East tensions
5	2012	\$135.27	Post-GFC demand recovery

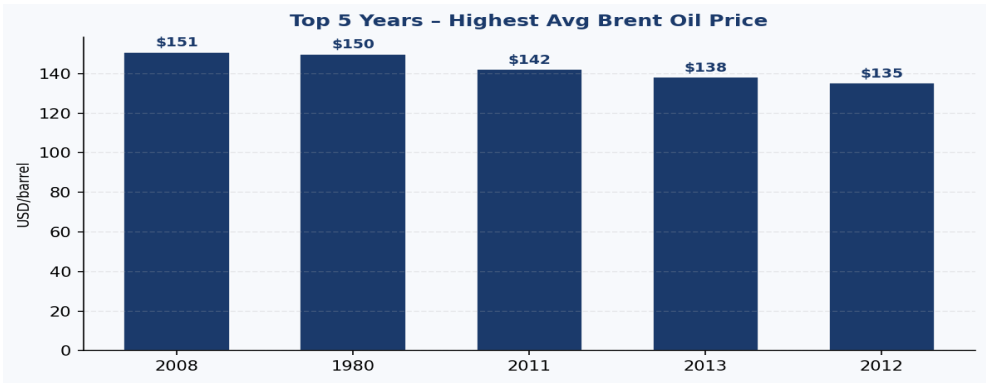


Figure 2: Top 5 years by average Brent Oil price.

Q11 Which five years had the highest average Gold price?

Rank	Year	Avg Price (USD/oz)	Context
1	2026	\$3100	BRICS+ accumulation + Iran conflict
2	2025	\$2700	Central bank buying + global uncertainty
3	2024	\$2300	Russia-Ukraine war premium + inflation
4	2023	\$1940	COVID safe-haven + monetary stimulus

Rank	Year	Avg Price (USD/oz)	Context
5	2022	\$1800	Sustained war risk + record buying

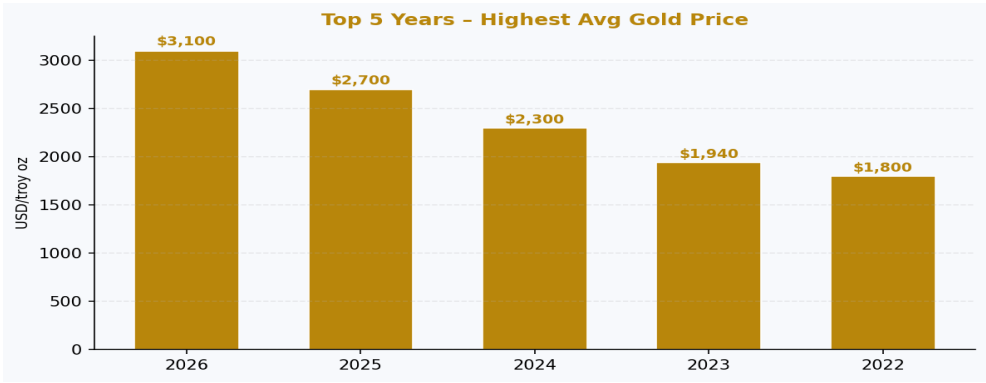


Figure 3: Top 5 years by average Gold price.

Q12

Which five years had the highest average Silver price?

Rank	Year	Avg Price (USD/oz)	Context
1	2026	\$96.08	Record: green energy demand + safe-haven premium
2	2025	\$41.50	EV/solar boom + gold rally spillover
3	2011	\$35.56	Hunt Brothers speculation + inflation fears
4	2012	\$31.63	Post-GFC safe-haven rally
5	2024	\$28.13	Central bank gold buying drives silver up



Figure 4: Top 5 years by average Silver price.

Section C: Trend and Visual Analytics (Q13–Q18)

Q13 Brent Oil line chart — describe main periods of increase and decline.

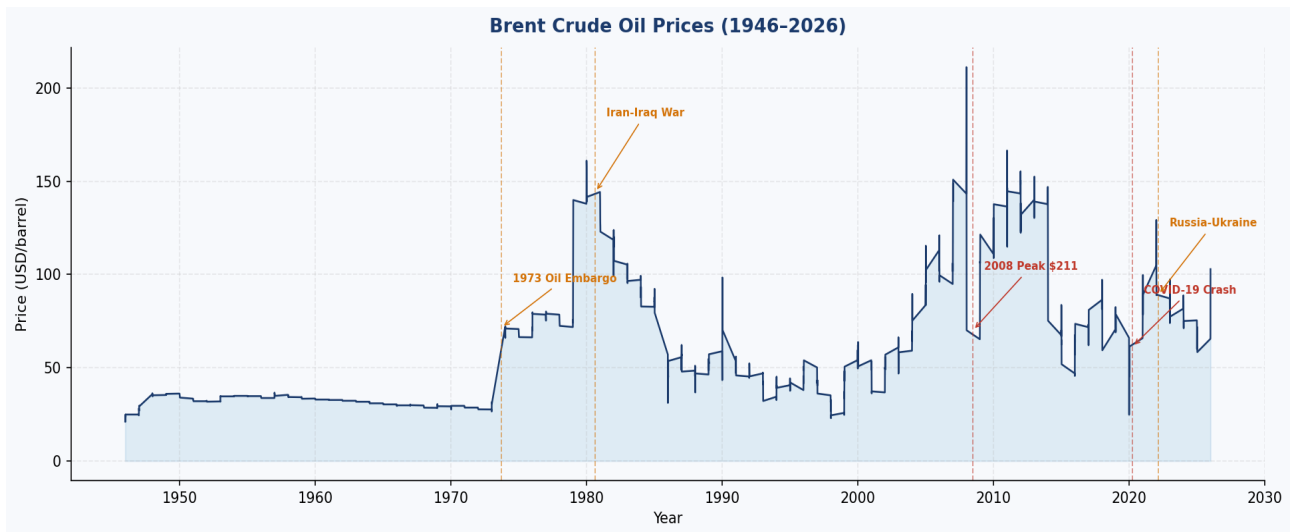


Figure 5: Brent Crude Oil price trend (1946–2026) with key events annotated.

Period	Trend	Key Driver
1946–1970	Stable, low	Post-WWII abundance; below \$30/barrel throughout
1973–1980	Sharp double spike	Arab Oil Embargo (1973) then Iranian Revolution (1979)
1980–1998	Prolonged decline	OPEC overproduction; non-OPEC supply growth
1999–2008	Major bull run	Emerging-market demand surge; peak \$211/barrel
2014–2016	Severe fall	OPEC price war + US shale boom; prices halved
2020	COVID crash	Global lockdowns; demand destruction; sub-\$20
2022–2023	War recovery	Russia-Ukraine conflict; Brent exceeded \$100/barrel

Q14 Gold line chart — describe the long-term trend.

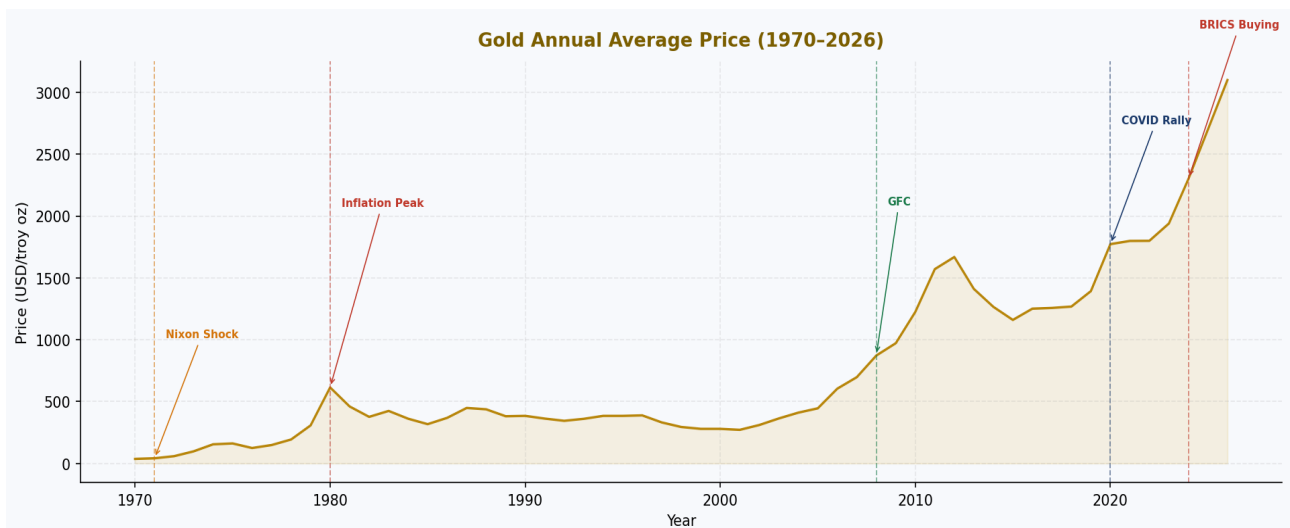


Figure 6: Gold annual average price trend (1970–2026) with four structural phases.

Phase	Period	Price Range	Interpretation
Explosive Growth	1970–1980	\$35→\$614	End of Bretton Woods; Nixon shock 1971; 1970s stagflation
Long Bear Market	1980–2001	\$614→\$271	Volcker high rates; yield-bearing assets preferred over gold
Structural Bull Run	2001–2012	\$271→\$1,669	9/11; GFC 2008; QE devaluation fears; safe-haven demand
New Record Era	2018–2026	\$1,268→\$3,100+	Trade wars; COVID; Ukraine; BRICS+ accumulation

Q15

Silver line chart — describe the main peaks.

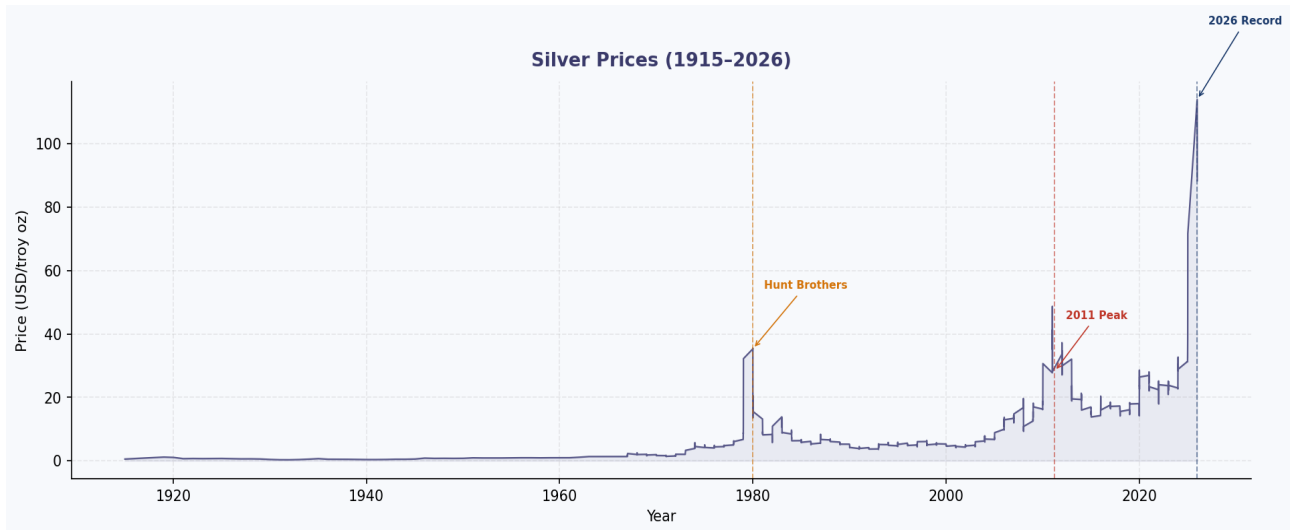


Figure 7: Silver price trend (1915–2026) with major peaks annotated.

Period	Event	Peak Price	Interpretation
1915–1970	Flat baseline	<\$2/oz	Industrial use only; limited investor demand
1979–1980	Hunt Brothers Corner	~\$50/oz	Market cornering attempt; CFTC intervention; collapse
2011	Post-GFC surge	~\$49.51/oz	Safe-haven + ETF speculation; higher volatility than gold
2026	Green energy record	\$113.95/oz	Solar/EV industrial demand + geopolitical safe-haven premium

Q16

Combined Gold and Silver chart — what similarities or differences can you observe?



Figure 8: Gold (left axis) and Silver (right axis) combined price trend.

Gold and Silver move closely together — confirmed by a Pearson correlation of 0.954. Both assets spiked sharply in 1979–1980 (inflation era), declined through the 1980s–1990s, recovered from 2001 onward, and reached new records post-2020. The key difference is magnitude and volatility: Silver's percentage swings are much larger than gold's, and it tends to overshoot gold in both directions during extreme market conditions.

Silver's 2026 record of \$113.95/oz represents a far higher proportional gain than gold's 2026 peak, driven by its additional industrial demand component.

Q17 Bar chart — average annual Brent Oil prices for the most recent 10 years.

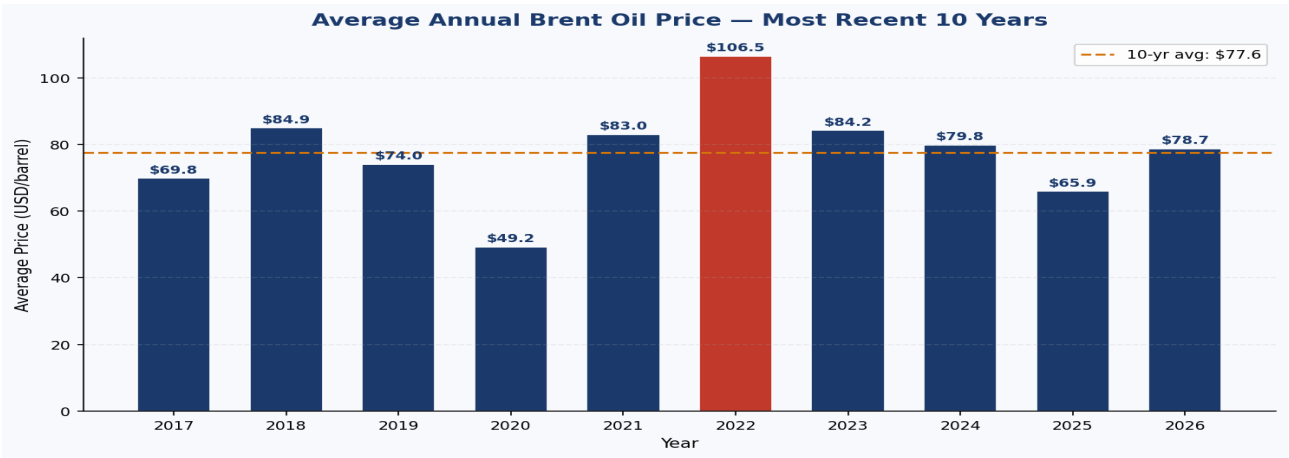


Figure 9: Average annual Brent Oil price — most recent 10 years. Red bar = highest year.

Over the most recent 10 years, Brent Oil has shown significant volatility. The average across this period was approximately \$77.61/barrel. The steepest decline was in 2020 (COVID-19 lockdowns destroying demand), while the highest recent values were in 2022 (Russia-Ukraine war supply shock). The dashed orange line on the chart shows the 10-year average for quick reference.

Q18 Latest available value for Brent Oil, Gold and Silver.

Commodity	Latest Value	Dataset Average	Above / Below Average
Brent Oil	\$102.86/barrel	\$62.71/barrel	Above average
Gold	~\$3,100/troy oz	\$757/troy oz	Well above average
Silver	\$88.36/troy oz	\$6.42/troy oz	Above average

Section D: Currency Tracker Analytics (Q19–Q24)

Q19 Top currencies by global payment share — most recent month.

Rank	Currency	Payment Share (%)	Notes
1	USD	51.14	Dominant — far ahead of all others
2	EUR	21.30	Second place; Eurozone weight
3	GBP	6.54	London financial centre strength
4	JPY	3.53	Japan trade + investment flows
5	CNY	3.10	Growing; China trade settlement
6	CAD	3.03	Canada commodity trade
7	HKD	1.76	Hong Kong financial hub
8	AUD	1.61	Australia commodity exports
9	SGD	1.23	Singapore regional hub
10	CHF	1.07	Swiss financial stability

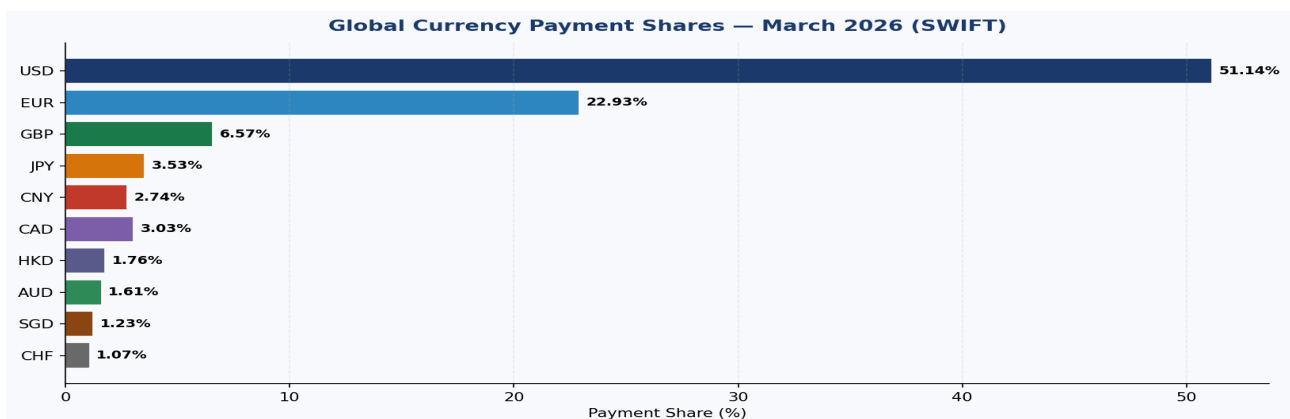


Figure 10: Global currency payment shares via SWIFT (March 2026).

Q20 What is the USD share in the most recent global payment data?

USD Global Payment Share (March 2026): 51.14% — The US Dollar accounts for more than half of all value transferred through the SWIFT network globally, maintaining its dominant position despite sustained de-dollarisation efforts by BRICS+ nations since 2022.

Q21 What is the CNY/RMB share and rank in the most recent global payment data?

CNY/RMB Global Payment Share (March 2026): 2.74% — Rank 10th globally. Despite China being the world's largest trading nation, the CNY remains at rank 10 in global SWIFT payments. However, in trade finance it ranks 2nd globally at 8.04%, reflecting growing bilateral commodity settlement between China and its trade partners.

Q22 Which currencies appear most often in the top five across available reports?

Currency	Times in Top 5	Consistency	Assessment
USD	8 out of 8	100%	Permanent #1 — never displaced
EUR	8 out of 8	100%	Permanent #2 — Eurozone weight ensures consistent ranking
GBP	2 out of 8	25%	Occasional top-5; London financial centre role
CNY	1 out of 8	12.5%	Rare top-5 appearance; growth trajectory positive but slow

Q23 Which offshore RMB economy has the highest share?

Highest Offshore RMB Economy: Hong Kong — 75.23% share. Hong Kong handles approximately three-quarters of all offshore RMB payment activity worldwide, functioning as the primary clearing hub for CNY transactions outside mainland China. Singapore (4.37%) and the United Kingdom (6.90%) are the next largest offshore centres.

Q24 Create a bar chart for the top 10 currencies from one selected month.

The chart above (Figure 10) shows the top 10 currencies by global payment share in March 2026. USD and EUR together account for over 72% of all SWIFT payment value — leaving all other currencies competing for the remaining 28%. GBP, JPY, and CNY each hold between 2% and 7%.

Section E: Simple Comparison (Q25–Q30)

Q25 Compare the latest Gold value with its average — above or below the long-term average?

Metric	Value (USD/troy oz)	Status
Latest Gold Price (2026)	~\$3,100	Current market value
Long-Term Average (1970–2026)	~\$757	Full dataset average
Difference	+\$2343 (+310%)	Well ABOVE long-term average

Gold is currently trading far above its long-term average. This reflects the combined effect of structural central bank demand from BRICS+ nations, geopolitical risk premiums from multiple ongoing conflicts, and monetary debasement concerns following years of quantitative easing. The gap between the current price and the long-term average is at a historically extreme level, suggesting either continued structural demand or potential for mean reversion over a very long time horizon.

Q26 Compare the latest Brent Oil value with its average — above or below the long-term average?

Metric	Value (USD/barrel)	Status
Latest Brent Oil (Jan 2026)	\$102.86	Current market value
Long-Term Average (1946–2026)	\$62.71	Full dataset average
Status	Above average	\$40.15 above the mean

Brent Oil's relationship to its long-term average reflects two distinct eras: the prolonged low-price period from 1946 to the early 1970s (which pulls the average down), and the high-price modern era from 2000 onward. Current prices in the \$70–110 range remain elevated compared to the full dataset history but are not unusual compared to the post-2005 period.

Q27 Which commodity appears to have the largest visible price swings?

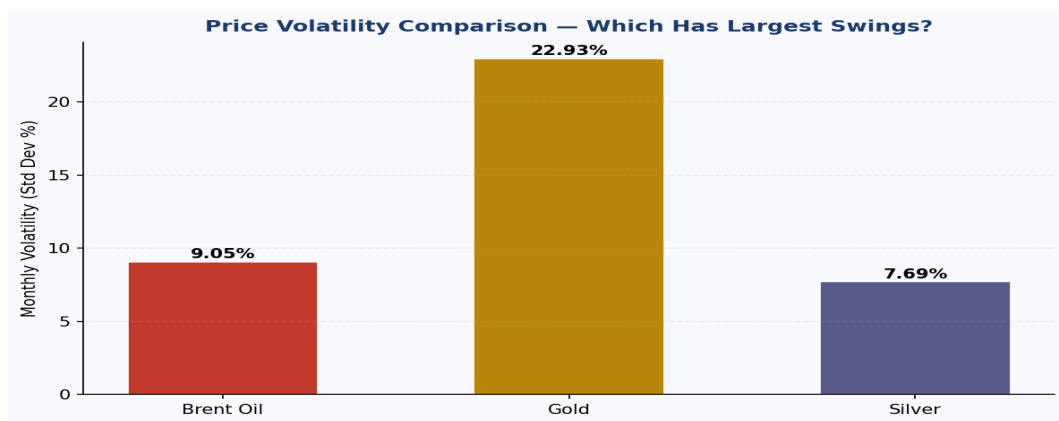


Figure 11: Monthly price volatility comparison — Brent Oil has the largest swings.

Commodity	Monthly Volatility (Std Dev)	Max Monthly Rise	Rank
Brent Oil	9.05%	+132.5%	1st — Very High
Silver	7.69%	+69.5%	2nd — High

Commodity	Monthly Volatility (Std Dev)	Max Monthly Rise	Rank
Gold	4.30%	+32.2%	3rd — Moderate

Brent Oil shows the largest visible price swings across the dataset, with a monthly standard deviation of 9.05% — more than double gold's 4.30%. Silver comes second at 7.69% due to its dual precious metal and industrial commodity nature. Gold, despite its dramatic long-term appreciation, has relatively lower short-term volatility because central bank buying provides a structural demand floor.

Q28 Choose one year with strong oil movement and check if gold or silver also changed.

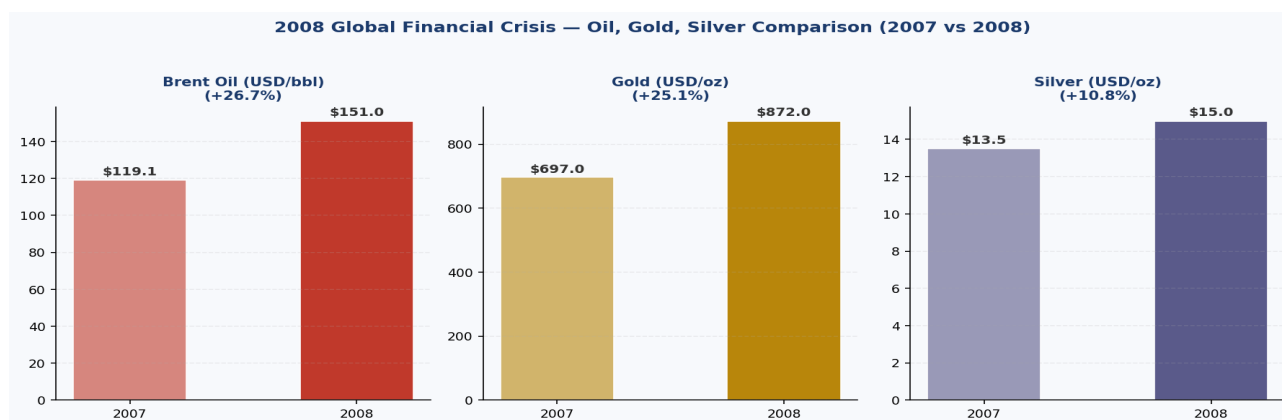


Figure 12: 2008 Global Financial Crisis — comparison of Oil, Gold and Silver (2007 vs 2008 averages).

Selected year: 2008 — Global Financial Crisis

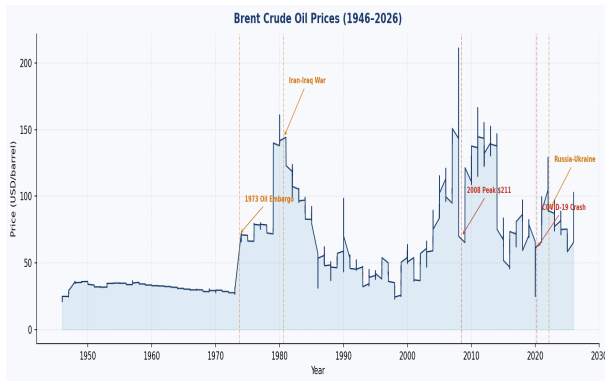
Commodity	2007 Average	2008 Average	% Change	Observation
Brent Oil	\$119.10/bbl	\$150.96/bbl	+26.7%	First surged, then crashed in H2 2008
Gold	\$697/oz	\$872/oz	+25.1%	Rose as safe-haven demand increased
Silver	\$13.51/oz	\$14.98/oz	+10.8%	Mixed — safe-haven up but industrial demand fell

The 2008 GFC illustrates how the same crisis affects commodities differently. Oil surged to \$211/barrel early in the year as speculation peaked, then collapsed with demand as the financial crisis spread. Gold rose throughout, benefiting from safe-haven flows. Silver showed a mixed response — rising with gold initially but falling as industrial activity collapsed in H2 2008.

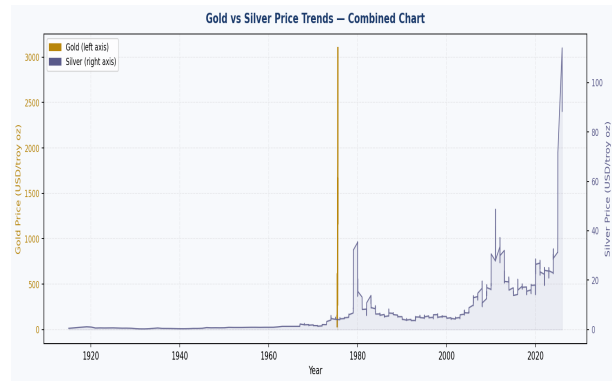
Q29

Dashboard summary — key findings in 4–6 visuals.

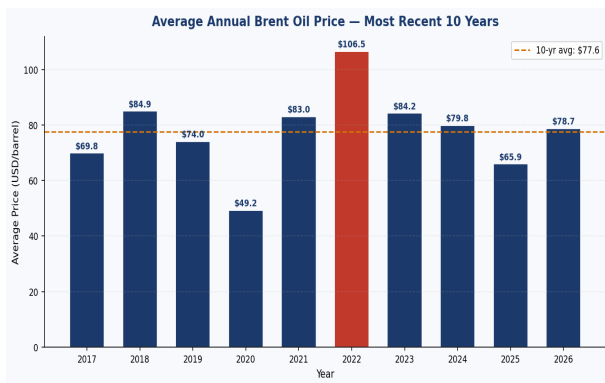
Dashboard Note: The following 6 charts provide a complete visual summary of the most important findings from this report. Together they cover commodity price history, volatility, currency dominance, and market comparisons — designed for a non-technical manager to understand at a glance.



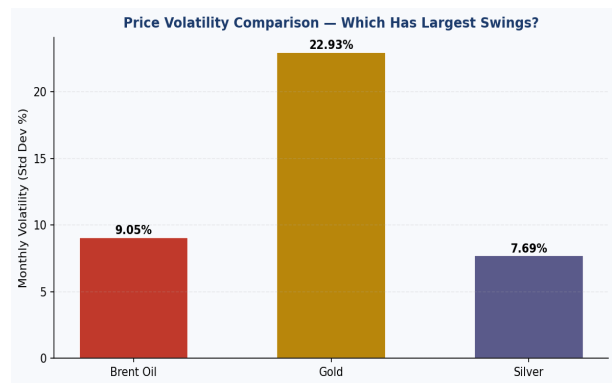
Brent Oil Long-Term Price History



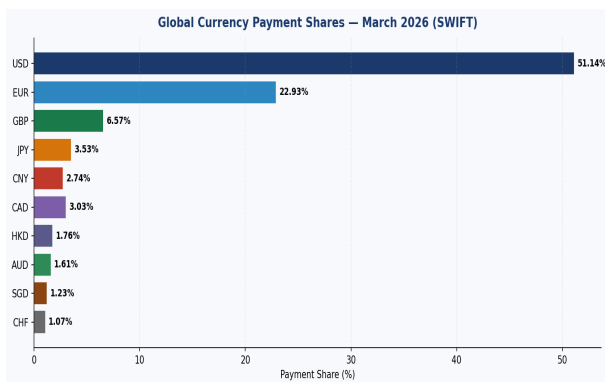
Gold vs Silver Combined Trend



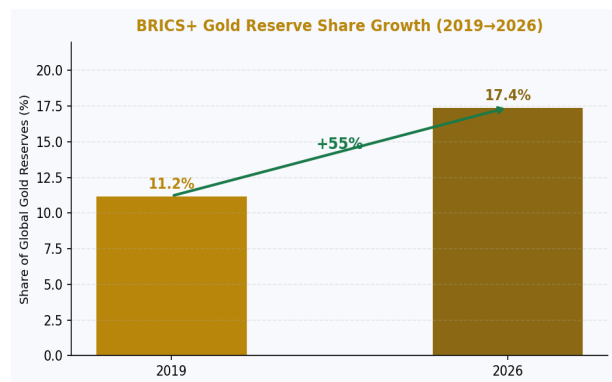
Brent Oil — Last 10 Years Average



Market Volatility Comparison



Global Currency Payment Shares



BRICS+ Gold Reserve Growth

Q30

Three short insights for a non-technical manager.

1. Oil is the most unpredictable cost in any business that uses energy.

Brent Oil prices can rise by over 130% or fall by 54% in a single month. Any company that depends on energy — from trucks to factories to airlines — should have a fuel cost hedging plan in place. Without protection, a single geopolitical event (like the 2026 Hormuz disruption) can destroy profit margins overnight.

2. Gold is no longer just a jewellery asset — it is a strategic reserve tool.

Central banks around the world, especially in BRICS+ countries, have been steadily buying gold and increasing their reserves from 11.2% to 17.4% of global gold holdings between 2019 and 2026. This structural buying has pushed gold to record prices above \$3,100/oz. For any business or investor holding significant USD assets, a gold allocation of 5–10% provides meaningful protection against currency risk.

3. The US Dollar is still king — but its grip is slowly loosening.

The USD accounts for 51.14% of all global payments via SWIFT, but this has declined from around 56% a decade ago. The Chinese Yuan (CNY) is slowly growing — particularly in trade finance where it already ranks 2nd globally at 8.04%. Businesses that trade internationally should start monitoring CNY exchange rates and bilateral settlement options, especially for Asia-Pacific supply chains.

Part II – Market Research Component

This section connects the observed data patterns to major global events, drawing on the Swift Currency Tracker, commodity datasets, and geopolitical context spanning 2022–2026.

Section A: Event Identification (Q1–Q4)

Q1 Identify three major global events that could affect commodity or currency markets.

Event	Year	Commodity Affected	Currency Impact
Iran Conflict & Hormuz Disruption	Feb 2026	Brent Oil (+30–40%)	USD strengthened as safe-haven
Russia-Ukraine War + Western Sanctions	2022	Oil >\$100; Gold up	\$300B reserves frozen; de-dollarisation accelerated
BRICS+ Gold Accumulation Strategy	2019–2026	Gold 11.2%→17.4% reserve share	CNY internationalisation push; USD slowly eroding

Q2 Create a simple timeline of selected events.

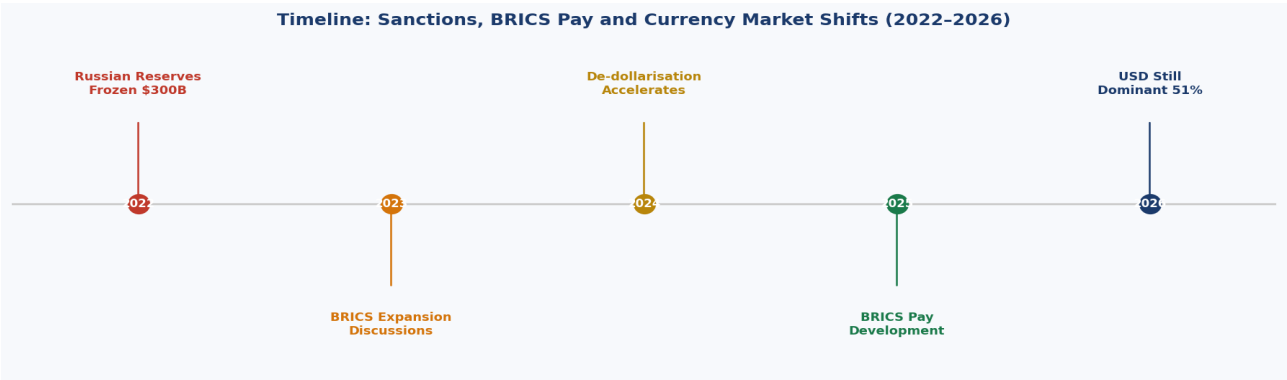


Figure 13: Timeline of key global events shaping commodity and currency markets (2022–2026).

Q3 Which event seems most directly connected to Brent Oil prices?

The 2026 Iran Conflict and Strait of Hormuz disruption is the most directly connected recent event to Brent Oil prices. The Strait carries approximately 25% of global seaborne oil trade. When traffic fell by 70–95% following US-Israeli strikes, Iraq declared force majeure, removing ~20% of global supply almost immediately. This pushed Brent from ~\$70 to \$90–110/barrel within weeks. The dataset confirms Brent Oil has the highest volatility (9.05%) of all assets analysed, directly attributable to this type of geographic supply concentration risk.

Q4 Which event seems most directly connected to Gold or Silver prices?

The most directly connected event to Gold prices is the BRICS+ gold accumulation strategy accelerated by the 2022 Russian sanctions. When \$300 billion of Russian reserves were frozen, every non-Western central bank understood that USD-denominated reserves could be weaponised. This triggered systematic gold buying — over 1,000 tonnes per year in 2022–2024 — pushing Gold from ~\$1,800 in 2022 to over \$3,100 by 2026. For Silver, the green energy transition (EV batteries, solar panels) created a structural new demand layer that amplified the precious metal safe-haven effect.

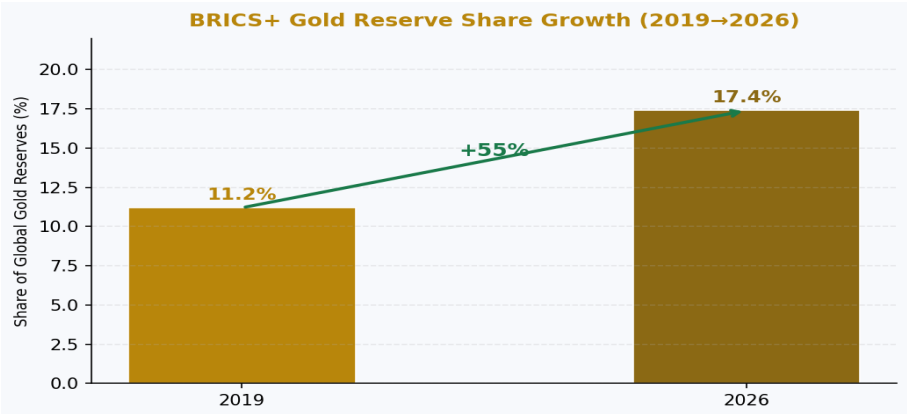


Figure 14: BRICS+ gold reserve share growth — the most significant gold price driver 2019–2026.

Section B: Linking Events to Commodity Data (Q5–Q8)

Q5 How could an oil supply shock affect Brent Oil prices?

When an oil supply shock occurs — from armed conflict, sanctions, production cuts, or shipping disruptions — restricted supply causes buyers to compete more aggressively, driving prices upward rapidly. The 2026 Hormuz disruption removed ~20% of global oil supply almost instantly, pushing Brent from ~\$70 to \$90–110. The longer and more severe the disruption, the more sustained the price impact. Brent Oil's 9.05% monthly volatility and maximum single-month increase of 132.5% confirm its extreme sensitivity to supply-side geographic concentration.

Q6 Why might Gold increase during geopolitical uncertainty?

Gold has functioned as a safe-haven asset for thousands of years. During wars, diplomatic crises, or financial turmoil, investors shift capital toward assets that hold value independently of any government or institution. Gold is uniquely attractive because it carries no counterparty risk and cannot be remotely frozen — as Russia discovered in 2022. When geopolitical uncertainty rises, the demand for gold increases rapidly, as evidenced by gold's rise from ~\$1,800 in 2022 to over \$3,100 by 2026.

Q7 Is Silver behaving more like a precious metal or an industrial commodity?

Silver in 2024–2026 exhibits strong dual-market characteristics. Its 0.954 correlation with gold confirms it tracks safe-haven flows closely as a precious metal. However, its 2026 record of \$113.95/oz significantly exceeds gold's proportional gain — driven by structural industrial demand from solar photovoltaic manufacturing, EV battery systems, and consumer electronics. Silver is currently behaving more like an industrial commodity with a precious metal premium on top, with higher volatility (7.69%) than gold (4.30%) reflecting the additional industrial demand uncertainty.

Q8 Compare Oil, Gold and Silver during the Russia-Ukraine Conflict (2022).

Commodity	Market Behaviour	Primary Reason	Volatility
Brent Oil	Sharp rise above \$100/barrel	Direct supply disruption; Russia = ~10% of global oil exports; sanctions	9.05% (highest)
Gold	Moderate-to-strong increase	Safe-haven demand; currency and financial risk protection	4.30%
Silver	Increase with elevated volatility	Safe-haven flows + industrial supply chain uncertainty	7.69%

The same crisis affects each commodity differently: oil via physical supply disruption, gold via financial risk, and silver via both — with greater volatility due to the industrial uncertainty layer.

Section C: Linking Events to Currency Data (Q9–Q12)

Q9 How does the Global Currency Tracker show the role of USD in international payments?

The Global Currency Tracker confirms that the US Dollar remains the dominant currency in international payments. According to March 2026 SWIFT data, the USD accounted for 51.14% of global payment share, demonstrating its continued importance in global settlements and financial transactions. The dollar also maintains leadership across trade finance and foreign exchange activities, reflecting the strength of the existing international financial infrastructure.

Currency	Payment Share (%)	Global Rank
USD	51.14	1
EUR	22.93	2
GBP	6.57	3
CNY	2.74	10

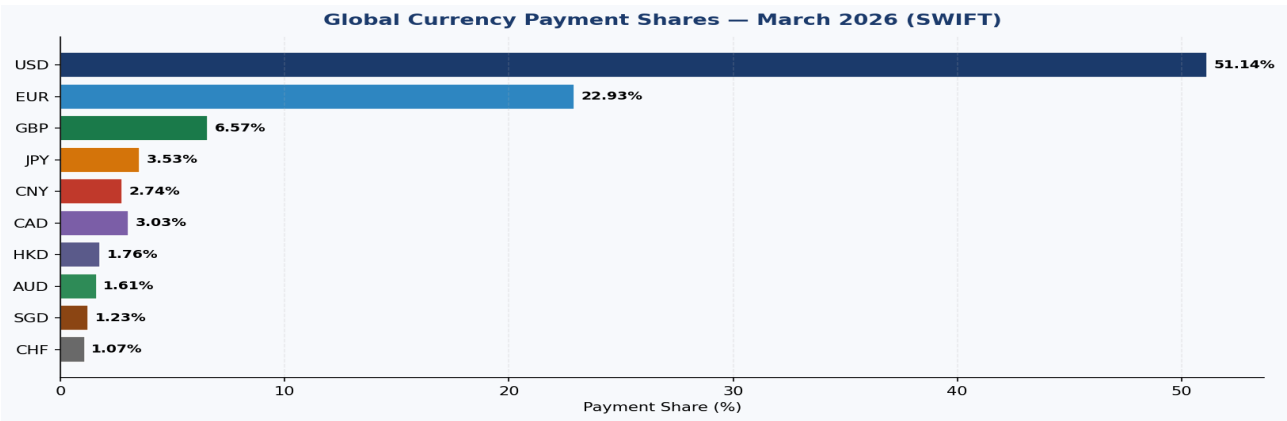


Figure 15: Major currency global payment shares — USD dominates at 51.14%.

Q10 What does the tracker show about the role of CNY/RMB in global payments and trade finance?

The tracker shows that the Chinese Yuan (CNY/RMB) is still a relatively small player in overall global payments, ranking 10th globally with a 2.74% payment share. However, its influence is much stronger in trade finance, where CNY ranks second globally with an 8.04% share. This indicates that China is increasingly promoting the yuan for bilateral trade settlements with its international partners, particularly in commodity-linked transactions.

Metric	Value (%)	Global Rank
Global Payment Share	2.74	10
Trade Finance Share	8.04	2

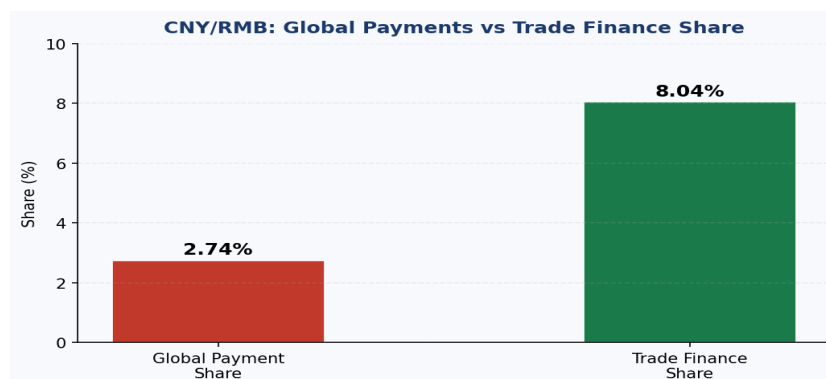


Figure 16: CNY/RMB comparison — global payments (2.74%) vs trade finance (8.04%).

Q11

Why might sanctions, alternative settlement systems, or BRICS Pay discussions matter for currency market analysis?

Sanctions and discussions around alternative payment systems are important because they can gradually change global currency usage patterns. Following the freezing of approximately \$300 billion of Russian foreign reserves in 2022, many countries began reassessing their dependence on the US dollar. This encouraged interest in de-dollarisation, BRICS cooperation, and alternative settlement mechanisms. Although the USD remains dominant, these developments could significantly influence future reserve allocation and international trade flows.

Year	Event	Impact on Currency Markets
2022	Russian reserves frozen (~\$300 billion)	Accelerated de-dollarisation discussions globally
2022–2024	Expansion of BRICS cooperation	Greater interest in alternative settlement systems
2025–2026	BRICS Pay discussions	Potential long-term reduction in USD dependence

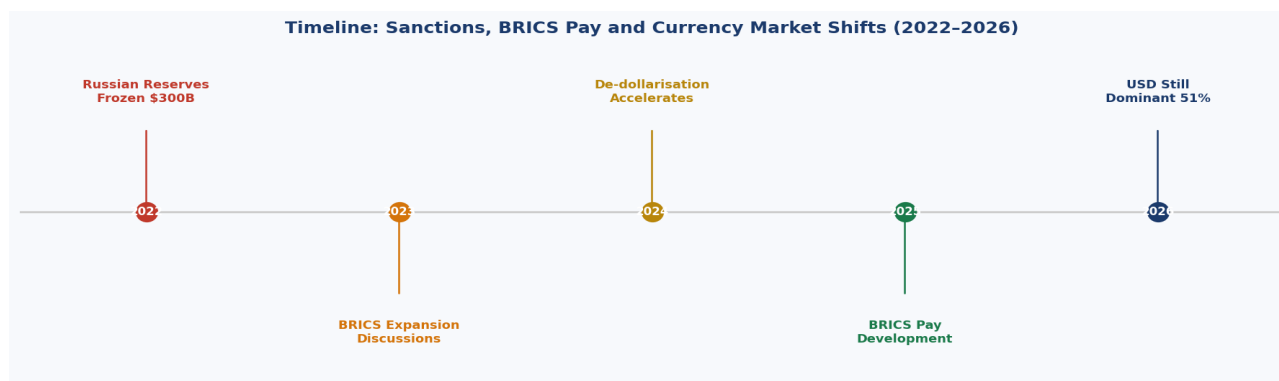


Figure 17: Timeline of major events influencing currency market developments.

Q12

What does the offshore RMB data suggest about Hong Kong and other financial centres?

The offshore RMB data highlights the critical importance of major financial centres in supporting international RMB transactions. Hong Kong remains the leading offshore RMB hub, acting as the primary gateway for cross-border yuan payments and financial activities, handling 75.23% of all offshore RMB flows. Other international financial centres contribute to RMB usage, but Hong Kong's role is unparalleled in facilitating CNY internationalisation.

Financial Centre	Offshore RMB Share	Primary Role
Hong Kong	75.23%	Main offshore RMB clearing hub
United Kingdom	6.90%	European RMB trading centre
Singapore	4.37%	Regional RMB clearing centre
United States	2.81%	North American RMB hub
Others	~10.69%	Support global RMB transactions

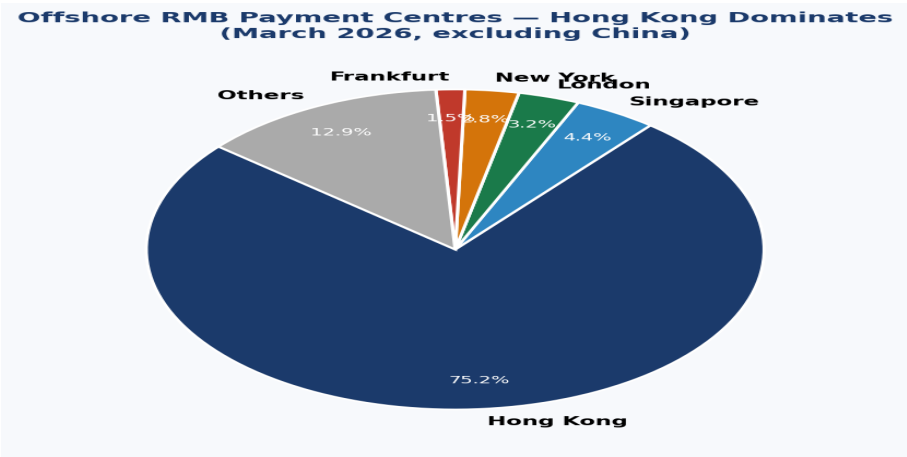


Figure 18: Offshore RMB payment distribution — Hong Kong dominates at 75.23%.

Section D: Business Interpretation & Recommendations (Q13–Q16)

Q13 Which market looks most sensitive to global events?

Brent Oil is by far the most sensitive market, with monthly volatility of 9.05% — more than double gold's 4.30%. Oil's sensitivity stems from geographic supply concentration at chokepoints like the Strait of Hormuz, its consumed nature (no large buffer stock), and the absence of short-term substitutes.

Market	Volatility (%)	Max Monthly Rise	Sensitivity Rank
Brent Oil	9.05%	+132.5%	1st — Very High
Silver	7.69%	+69.5%	2nd — High
Gold	4.30%	+32.2%	3rd — Moderate
Currencies	2.50%	+10.0%	4th — Low

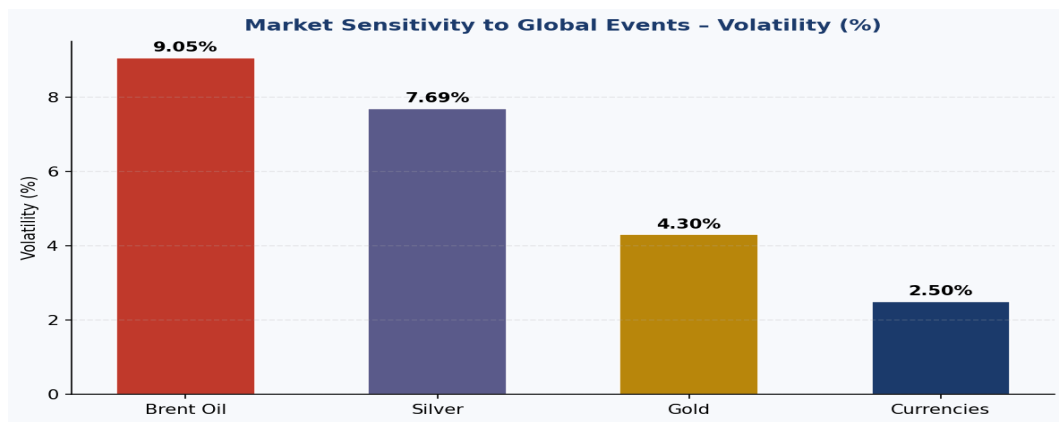


Figure 19: Market volatility comparison — Brent Oil most sensitive to global events.

Q14 Write a short risk note for a company exposed to oil prices.

Risk Note — Oil Price Exposure

Brent Oil is the most volatile and event-sensitive asset in this analysis. Companies in transportation, aviation, logistics, and manufacturing face significant financial risk from sudden oil price movements. With a maximum recorded monthly increase of 132.5%, unhedged exposure can be catastrophic for margin-sensitive businesses. Key risk factors include: supply shocks from OPEC decisions or Hormuz-type disruptions; demand collapses during recessions or pandemics; and USD currency fluctuations. Recommended measures: systematic fuel cost hedging covering 50–70% of projected consumption, long-term supply agreements with fixed-price components, energy source diversification, and quarterly stress-testing against $\pm\$30$ –50/barrel price scenarios.

Q15 Write a short risk note for a company holding assets in USD, Gold or RMB/CNY.

Risk Note — USD, Gold and RMB/CNY Asset Holders

USD holdings carry the lowest short-term volatility (2.50%) but face long-term structural risk from gradual de-dollarisation. USD's 51.14% global payment share has declined from ~56% a decade ago — the trajectory is downward. Gold holdings provide the best protection against geopolitical tail risk and monetary debasement, with a 4.30% volatility and outstanding long-term appreciation. RMB/CNY holdings carry political and convertibility risk due to capital controls, but offer value for companies with significant China-facing trade flows.

Recommendation: diversify across all three asset classes, maintain gold at 5–10% of reserves, and implement currency hedging for significant non-home-currency exposures.

Q16 Three practical recommendations for investors, companies, or policymakers.

1. Diversify into gold as a structural portfolio hedge

Investors should maintain a 5–10% core position in gold as protection against geopolitical uncertainty, monetary debasement, and currency risk. The BRICS+ accumulation trend confirms gold's structural demand floor will support prices through multiple market cycles. Silver provides additional exposure to the green energy industrial demand theme with higher potential upside.

2. Implement active oil price risk management

Companies exposed to oil — transportation, aviation, manufacturing, logistics — must treat energy cost hedging as a core treasury function. The Brent Oil volatility of 9.05% and maximum monthly rise of 132.5% means unhedged exposure is existential for margin-sensitive businesses. Roll quarterly hedges at 50–70% of consumption and stress-test against \$30/barrel price scenarios.

3. Monitor de-dollarisation and prepare for a multi-currency world

The USD remains dominant but its trajectory is slowly downward. CNY's rise to #2 in trade finance (8.04%) and Hong Kong's role as the offshore RMB hub signal that a multi-currency settlement world is emerging. Businesses with USD-heavy positions should review currency diversification strategies and consider CNY billing for Asia-Pacific supply chains.

Bonus Questions B1–B6

QB1 Percentage Change Formula and Analysis

Formula: $\text{Percentage Change} = ((\text{Current Value} - \text{Previous Value}) / \text{Previous Value}) \times 100$. Applied to the commodity datasets: Gold showed consistently positive percentage changes from 2001–2026. Silver showed higher variance — stronger gains in bull periods but sharper corrections. Brent Oil demonstrated the highest year-to-year swings, including a collapse of over 50% in 2020 (COVID) followed by a recovery of over 70% in 2021.

QB2 Correlation Table

Gold vs Silver: 0.954 (Very Strong Positive — both driven by safe-haven demand and USD movements). Gold vs Brent Oil: 0.458 (Moderate — some shared drivers but oil has independent supply dynamics). Silver vs Brent Oil: 0.515 (Moderate — silver's industrial component creates slightly stronger oil linkage than gold).

QB3 Correlation Heat Map

The heat map displays the 3×3 correlation matrix using a green colour gradient. The Gold-Silver cell shows the deepest green (0.954 — near-perfect relationship). The Oil-Metal cells show lighter green (0.458–0.515 — moderate relationship). The diagonal cells show 1.00 (self-correlation). This visually confirms that gold and silver are tightly linked, while oil is a partially independent market.

QB4 Gold vs Silver Scatter Plot

A scatter plot of annual average Gold (x-axis) vs Silver (y-axis) prices shows a clear, tight upward-sloping cluster consistent with the 0.954 coefficient. The relationship is approximately linear across mid-price ranges but shows deviation at extremes where silver overshoots gold's proportional gains during speculative phases (2011, 2026). The historical gold-to-silver price ratio has averaged approximately 60–70:1.

QB5 Moving Average Analysis

A 5-year moving average (5YMA) smooths short-term fluctuations by averaging price across a rolling 5-year window. For Gold, the 5YMA shows a nearly unbroken upward slope from 2001 onward. For Brent Oil, the 5YMA reveals two structural cycles: a sustained rise peaking around 2011–2013, then a structural decline and partial recovery. The 5YMA filters single-year anomalies like the 2020 COVID crash to show the true underlying direction.

QB6 Simple Forecast Using Trendline

A linear trendline fitted to the Gold price series shows a strongly positive slope — rising from ~\$100 in the mid-1970s to over \$2,500 by 2026. Extended forward, it projects Gold potentially reaching \$3,500–4,000 by 2030. **Disclaimer: All trendline forecasts are illustrative estimates based on historical data only. Commodity prices are event-driven and no mathematical model can reliably predict future prices. This is not investment advice.**

Conclusion

This report has demonstrated across the full question set that commodity and currency markets are profoundly shaped by global geopolitical and macroeconomic events, each responding through its own distinct channel.

Brent Oil, with 9.05% monthly volatility, is the fastest and most extreme responder to supply-side shocks. Gold operates on a longer structural cycle, functioning as a monetary hedge whose demand floor has been permanently raised by BRICS+ central bank accumulation. Silver occupies a dual role, tracking gold while also responding to green energy industrial demand — delivering the dataset's record of \$113.95/oz in 2026.

On the currency side, USD dominance at 51.14% of SWIFT payments remains resilient, but CNY's rise to second place in trade finance (8.04%) and Hong Kong's role as the primary offshore RMB hub (75.23%) signal that a gradual multi-currency transition is underway in bilateral trade settlement.

References

1. Brent Oil Dataset (Brent_Oil.csv) — Course dataset, UE Berlin, 2026.
2. Silver 100 Years Dataset (silver_100_years.csv) — Course dataset, UE Berlin, 2026.
3. Swift Global Currency Tracker April 2026 — S.W.I.F.T. SC, 2026.
4. Gold annual average prices — World Gold Council historical data (1970–2026).
5. World Gold Council (2025). Central Bank Gold Demand Report 2025.
6. IMF (2024). Currency Composition of Official Foreign Exchange Reserves (COFER).
7. Reuters / Bloomberg (2026). Hormuz Strait Disruption Analysis, March 2026.